Question	Answer	Marks
7(a)	(magnetic) force (always) normal to velocity/direction of motion	**M1**
	(magnitude of magnetic) force constant **or** speed is constant/kinetic energy is constant	**M1**
	so provides the centripetal force	**A1**
7(b)	increase in KE = loss in PE **or** $\frac{1}{2} mv^2 = qV$	**M1**
	$p = mv$ with algebra leading to $p = \sqrt{(2mqV)}$	**A1**
7(c)	$Bqv = mv^2 / r$ $mv = Bqr$ **or** $p = Bqr$	**C1**
	$(2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 120)^{1/2} = B \times 1.60 \times 10^{-19} \times 0.074$	**C1**
	$B = 5.0 \times 10^{-4}$ T	**A1**
7(d)	greater momentum	**M1**
	($p = Bqr$ and) so r increased	**A1**

Question	Answer	Marks
8	strong (uniform) magnetic field	B1